

Recap:

we have seen three types of convergence of Fourier series

1. Pointwise: Assume that x is of bounded variation
$$S_N(t_0) \rightarrow \begin{cases} x(t_0) & \text{if } x \text{ is continuous at } t_0 \\ \frac{x(t_0^+) + x(t_0^-)}{2}, & \text{if not} \end{cases}$$

2. Mean square: If x is any Riemann integrable signal:

$$\|S_N - x\|^2 \rightarrow 0.$$

3. Uniform: If x ??:

$$\|S_N - x\|_\infty \rightarrow 0 \text{ as } N \rightarrow \infty.$$

I. A sufficient condition for uniform convergence

Theorem: If x is continuous at all time t , and
if its Fourier series coefficients satisfy:

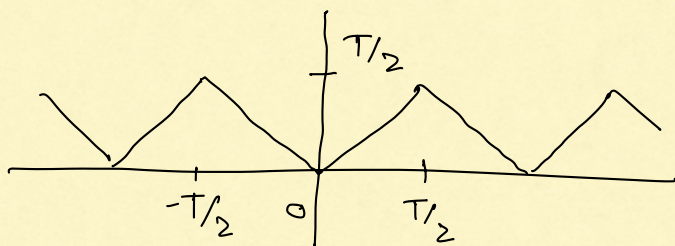
$$\sum_{n=-\infty}^{+\infty} |c_n| \text{ is finite}$$

then:

$$S_N \rightarrow x \text{ uniformly.}$$

Example:

① Triangular wave



$$c_n = \begin{cases} -\frac{2}{\pi n^2} & \text{if } n \text{ is odd} \\ \pi/2 & \text{if } n=0 \\ 0 & \text{otherwise} \end{cases}$$

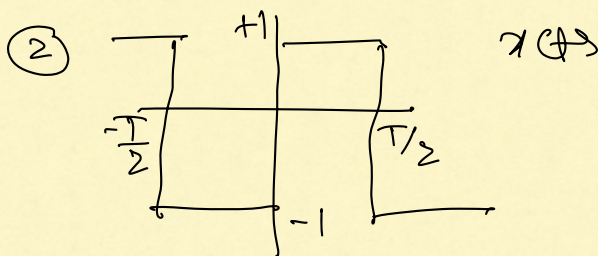
$$\sum_{n=-\infty}^{+\infty} |c_n| = \frac{\pi}{2} + \sum_{\substack{n \geq 1 \\ \text{odd}}} 2 \times \left| -\frac{2}{\pi n^2} \right|$$

$$= \frac{\pi}{2} + \sum_{\substack{n \geq 1 \\ \text{odd}}} \frac{4}{\pi n^2}$$

$$\leq \frac{\pi}{2} + \sum_{n \geq 1} \frac{4}{\pi n^2}$$

$$= \frac{\pi}{2} + \frac{4}{\pi^2} \times \frac{\pi^2}{6} < \infty.$$

$\therefore$ For this signal, $S_N \rightarrow x$ uniformly.



$$c_n = \begin{cases} 0, & n=0 \\ \frac{-1 + (-1)^n}{j\pi n}, & n \neq 0. \end{cases}$$

$$c_n = \begin{cases} 0, & n \text{ is even} \\ -\frac{2}{j\pi n}, & n \text{ odd.} \end{cases}$$

$$|c_n| = \begin{cases} 0, & n \text{ even} \\ \frac{2/\pi}{n}, & n \text{ odd} \end{cases}$$

$$\sum_{n=-\infty}^{+\infty} |c_n| = 2 \times \sum_{\substack{n \geq 1 \\ \text{odd}}} \frac{2}{\pi} \cdot \frac{1}{n} = \frac{4}{\pi} \sum_{\substack{n \geq 1 \\ \text{odd}}} \frac{1}{n}$$

This does not converge

But Parseval's still holds since $\sum |c_n|^2 < \infty$.

II. Gibb's Phenomenon:

What if a signal is of bounded variation but it has a discontinuity at some point t_c .
(like the square wave examples).

In such a case, we know from Dirichlet-Jordan that pointwise convergence holds.

But what about uniform convergence?

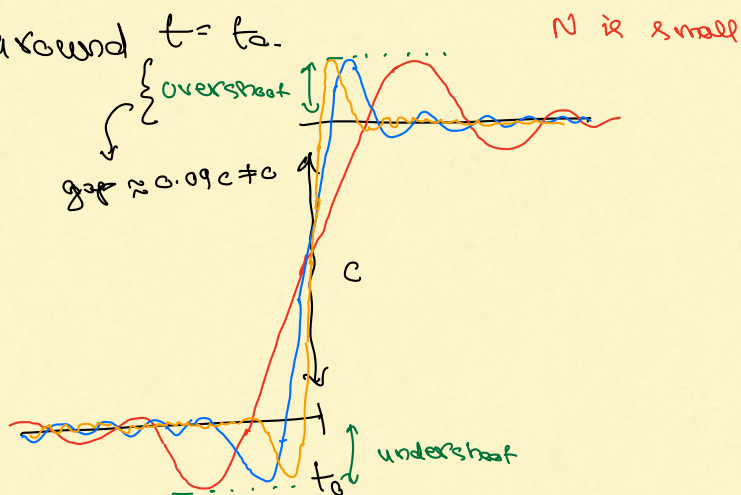
The answer is no. We do not have uniform convergence for a signal with discontinuity.

In particular, if there is a discontinuity at t_c ,

$$\text{let } c = |x(t_c^+) - x(t_c^-)|$$

As $N \rightarrow \infty$, the partial sum S_N overshoots/undershoots x by approximately $0.09c$

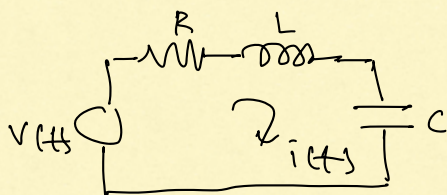
around $t = t_c$.



If x has a discontinuity $\Rightarrow$ Uniform convergence does NOT happen.

III. Second-order RLC circuit

Ref: EE1101, Lecture 28-29.



$$v(t) = R i(t) + L \frac{di}{dt} + \frac{1}{C} \int i(t) dt.$$

$$\frac{dv}{dt} = R \frac{di}{dt} + L \frac{d^2 i}{dt^2} + \frac{1}{C} i$$

$$\frac{1}{L} \frac{dv}{dt} = \frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i$$

$$\text{If } v(t) = e^{j\omega t}, \quad i(0) = u, \quad i'(0) = v.$$

$$\hat{i}(t) = \hat{i}_c(t) + \hat{i}_p(t)$$

↖
Complementary
Soln.

↗ Particular
Soln.
or
Steady-State Soln.

$$\begin{aligned} \textcircled{1} \quad \hat{i}_p(t) &= \gamma \cdot e^{j(\omega t + \theta)} = \gamma e^{j\theta} \cdot e^{j\omega t} \\ &= z \cdot e^{j\omega t}, \quad z \in \mathbb{C}. \end{aligned}$$

$$\textcircled{2} \quad \hat{i}_c(t) \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

Suppose we are interested only in the steady-state soln.

Let us abstract the circuit analysis equation:

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = \frac{1}{L} \frac{dv}{dt}.$$

as the differential equation

$$x'' + 2k \cdot x' + b x = f$$

Given: above eqn., f, k, b

we will assume that $k, b > 0$ ($k, b \in \mathbb{R}$)

f is continuous.

Ref: Chap 23 of

Körner, "Fourier Analysis".